

BASRAH INTL, Iraq

WMO: 406890

Lat: 30.549N

Lon: 47.662E

Elev: 3

StdP: 101.28

Time Zone: 3.00 (E03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	4.8	-7.2	2.0	18.1	-5.2	2.4	18.5	14.5	18.6	12.4	17.4	2.7	270	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	49.1	21.5	48.0	21.5	46.9	21.3	26.7	35.4	24.7	35.4	23.2	37.2	6.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	19.3	31.6	21.9	16.6	30.1	19.8	14.5	28.2	84.0	35.1	75.2	34.4	69.1	36.5	33.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.3	11.5	9.9	-0.3	50.9	1.4	1.7	-1.4	52.1	-2.2	53.0	-3.0	54.0	-4.1	55.2		
			-1.9	29.0	1.5	1.8	-3.0	30.3	-3.8	31.4	-4.7	32.4	-5.7	33.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	13.1	15.7	20.8	26.3	33.0	36.4	37.9	37.5	33.6	28.2	19.1	14.0
	DBStd	9.56	3.09	3.48	3.50	3.38	2.99	2.22	2.00	1.89	2.52	3.27	4.15	3.43
	HDD10.0	16	8	3	0	0	0	0	0	0	0	0	0	6
	HDD18.3	442	162	87	15	1	0	0	0	0	0	0	39	138
	CDD10.0	5993	105	161	336	490	712	793	866	852	707	565	274	131
	CDD18.3	3377	1	12	93	241	454	543	608	594	457	307	63	5
	CDH23.3	55086	6	98	925	2924	7357	9701	11065	10631	7513	4166	664	34
	CDH26.7	39252	0	13	342	1533	5059	7337	8594	8178	5406	2555	233	2
(14) Wind	WSAvg	4.4	3.7	4.1	4.5	4.5	4.6	6.0	5.7	4.7	4.0	3.6	3.5	3.5
(15) Precipitation	PrecAvg	315	67	42	47	29	7	1	1	0	1	12	45	63
	PrecMax	453	164	101	149	102	47	3	2	1	7	56	192	173
	PrecMin	170	12	1	5	1	0	0	0	0	0	1	1	5
	PrecStd	82	37	25	33	23	9	1	1	0	1	15	46	44
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	28.2	37.0	40.1	46.1	49.2	50.9	50.2	48.0	42.8	33.9	26.1
		MCWB	15.0	15.4	16.5	19.0	20.9	21.1	21.6	21.6	21.4	19.7	18.0	15.0
	2%	DB	22.0	26.0	32.2	37.8	44.1	47.8	49.2	49.0	46.1	40.2	31.2	24.1
		MCWB	14.0	14.6	16.3	18.5	20.2	20.9	21.4	21.8	21.1	19.4	17.5	15.2
	5%	DB	20.8	24.1	30.1	35.8	42.2	46.4	48.1	47.9	44.8	38.9	29.2	22.2
		MCWB	13.7	13.8	15.6	17.9	19.6	20.7	21.4	21.8	20.6	19.2	16.9	14.8
	10%	DB	19.1	22.3	28.1	33.9	40.9	45.1	46.9	46.8	43.2	37.2	27.2	20.8
		MCWB	13.0	13.4	14.9	17.4	19.4	20.3	21.2	21.6	20.2	18.6	17.1	14.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.4	18.2	20.2	21.5	23.2	25.8	27.6	29.4	28.2	26.4	23.4	19.6
		MCDB	20.6	23.0	25.2	30.2	34.0	36.9	35.6	34.2	32.6	31.5	27.5	22.7
	2%	WB	17.0	17.1	18.9	20.5	22.0	22.7	25.2	27.2	24.8	24.7	21.4	18.0
		MCDB	19.5	21.6	25.4	30.3	34.7	41.2	38.0	36.0	35.6	29.7	25.9	21.2
	5%	WB	15.6	16.0	17.8	19.7	21.2	21.6	23.0	24.6	22.7	23.4	19.7	16.4
		MCDB	18.3	20.7	24.8	29.7	35.8	41.8	42.5	39.7	38.5	29.2	24.7	19.8
	10%	WB	14.3	14.9	16.8	19.1	20.6	21.0	22.0	22.7	21.3	21.7	18.5	15.2
		MCDB	17.5	19.9	25.0	29.3	35.9	42.4	43.9	42.7	40.1	29.7	24.3	18.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.1	13.2	13.2	14.3	17.2	17.7	18.7	19.4	16.9	13.7	11.9
		MCDBR	12.3	13.9	15.5	15.1	16.1	18.2	19.1	19.7	20.3	18.4	15.6	13.5
		MCWBR	6.7	5.8	5.2	4.8	4.9	5.1	5.6	6.4	6.3	7.1	7.2	7.2
	5% WB	MCDBR	8.7	10.1	11.8	12.3	14.2	17.2	18.1	17.9	18.7	13.8	10.7	9.4
		MCWBR	5.9	5.6	5.3	4.7	4.7	5.3	6.4	8.0	7.3	7.4	6.3	6.4
(40) Clear-Sky Solar Irradiance	taub		0.494	0.532	0.591	0.686	0.679	0.612	0.656	0.605	0.549	0.577	0.531	0.498
	taud		1.844	1.761	1.632	1.510	1.522	1.608	1.552	1.641	1.742	1.701	1.787	1.847
	Ebn at Noon		698	715	707	662	671	715	681	710	732	665	654	666
	Edh at Noon		172	205	249	291	289	264	278	252	219	214	181	164
(44) All-Sky Solar Radiation	RadAvg		3.42	4.38	5.56	6.13	7.10	8.05	7.61	7.14	6.28	4.84	3.65	3.17
	RadStd		0.23	0.25	0.22	0.33	0.34	0.22	0.22	0.18	0.21	0.21	0.26	0.29

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.29	N/A	N/A	+0.44	N/A	N/A	N/A	N/A	+85	+83

Nomenclature: See separate page